

B.Sc. (Hons.) in Environmental Science

- Environmental Science is a multidisciplinary field that combines natural sciences, social sciences, and other disciplines to study the environment and the way humans interact with it.
- It focuses on the understanding of processes and systems that operate in the natural environment, as well as the impact of human activities on the environment and its resources.
- Students learn to evaluate the impacts of human activities and develop solutions to address environmental issues.
- Courses cover topics such as ecology, geology, hydrology, atmospheric science, oceanography, biology, chemistry, and physics.
- Students also learn about environmental law, policy, economics, and management, as well as the principles of sustainability.
- Graduates of this program are prepared to work in a variety of fields, including environmental consulting, natural resource management, and environmental education.

Unit 1 - Environment:

- Definition: The environment is a complex web of physical, biological, and chemical factors that interact and influence the living and nonliving components of the planet.
- Types of Environment: Terrestrial, Aquatic, and Marine
- Components of environment: Atmosphere, Hydrosphere, Lithosphere, and Biosphere
- Segments of environment: Air, Water, Land, Flora, and Fauna
- Scope and importance: The environment is essential for the survival of all living things on the planet. It provides the resources necessary for life, such as food, water, and shelter. It also provides the means for the circulation of energy, and it helps to regulate the climate.
- Need for Public Awareness: Public awareness of environmental issues is essential for the protection and preservation of the environment. It is important for people to understand the impacts of their actions on the environment and to take steps to reduce their environmental footprint.

Ecosystem:

- Definition: An ecosystem is a complex network of living and nonliving components that interact and influence each other.
- Types of Ecosystem: Marine, Terrestrial, and Aquatic
- Structure of Ecosystem: An ecosystem is made up of four main components: abiotic factors, producers, consumers, and decomposers.
- Food Chain: A food chain is a linear sequence of organisms that are linked by their feeding relationships.
- Food Web: A food web is a complex network of interconnected food chains.

- Ecological Pyramid: An ecological pyramid is a graphical representation of the trophic levels in an ecosystem.
- Balance Ecosystem: A balanced ecosystem is one in which all the components of the system are in equilibrium and functioning properly.

Effects of

Unit 2 - Natural Resources: Introduction, Classification

1. Water Resources:
 - Availability, sources and Quality Aspects
 - Water Borne and Water Induced Diseases
 - Fluoride and Arsenic Problems in Drinking Water
2. Mineral Resources:
 - Material Cycles
 - Carbon, Nitrogen and Sulfur cycles
3. Energy Resources:
 - Conventional and Non conventional Sources of Energy
4. Forest Resources:
 - Availability
 - Depletion of Forests
 - Environment impact of forest depletion on society

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid Waste Management

- Pollution is the introduction of contaminants into the environment that cause harm or discomfort to humans and other living organisms.
- Air pollution is the release of harmful chemicals into the atmosphere, such as carbon dioxide, sulfur dioxide, and nitrogen oxides. It can cause respiratory illnesses, heart disease, and even death.
- Water pollution is the contamination of water bodies, such as lakes, rivers, oceans, and groundwater, with harmful substances. It can lead to the spread of disease, the destruction of aquatic life, and the contamination of drinking water.
- Soil pollution is the contamination of soil with toxic chemicals, such as pesticides, fertilizers, and industrial waste. It can lead to the destruction of crops and the contamination of groundwater.
- Noise pollution is the excessive noise caused by human activities, such as construction, traffic, and industrial operations. It can cause hearing loss, stress, and sleep disturbances.
- Solid waste management is the process of collecting, treating, and disposing of solid waste in a safe and efficient manner. It includes the collection, transportation, and disposal of solid waste, as well as the recycling of materials for reuse.

Unit 4 - Current Environmental Issues of Importance

- **Global Warming:** Global warming is the long-term rise in the average temperature of the Earth's climate system. It is a major aspect of climate change, and has been demonstrated by direct temperature measurements and by measurements of various effects of the warming.
- **Green House Effects:** Greenhouse effect is the process by which radiation from a planet's atmosphere warms the planet's surface to a temperature above what it would be without its atmosphere.
- **Climate Change:** Climate change is a long-term shift in the average weather patterns that characterize a particular region. It is caused by changes in the atmosphere, the land surface, and the ocean, and is linked to both natural and human activities.
- **Acid Rain:** Acid rain is rain that has been made acidic by certain pollutants in the air. It can have harmful effects on plants, aquatic animals, and infrastructure.
- **Ozone Layer Formation and Depletion:** The ozone layer is a region of the Earth's atmosphere that absorbs most of the Sun's ultraviolet radiation. It is being depleted by certain pollutants, leading to increased levels of ultraviolet radiation reaching the Earth's surface.
- **Population Growth and Automobile Pollution:** Population growth and automobile pollution are two of the major causes of air pollution. Automobile pollution is caused by the burning of fossil fuels, and population growth can lead to increased emissions of pollutants.
- **Burning of Paddy Straw:** Burning of paddy straw is a major source of air pollution in many parts of the world. Paddy straw burning releases carbon dioxide, nitrogen oxides, and other pollutants into the air, which can have a detrimental effect on human health and the environment.

Unit 5 - Environmental Protection

Environmental Protection Act 1986

The Environmental Protection Act 1986 is an Act of Parliament in the United Kingdom that provides a framework for environmental protection in the UK. The Act sets out the principles of environmental protection and the duties of public bodies and other organisations in relation to the environment. It also provides for the setting up of a system of environmental protection notices and orders, and for the enforcement of those notices and orders.

Initiatives by Non Governmental Organizations (NGO's)

Non-governmental organisations (NGOs) are organisations that are independent of government, and which are often involved in the promotion of environmental protection and sustainability. NGOs often work to raise awareness of environmental issues, and to lobby governments and other organisations to take action to protect the environment. Examples of NGOs that work on environmental

protection include Greenpeace, Friends of the Earth, and the World Wildlife Fund.